

Coloring of planar graphs

Introduction

An algorithm for the coloring of planar graphs for multiple **coloring variants** is proposed. These coloring variants include:

- Proper coloring
- Two **adjacent** vertices have different colors

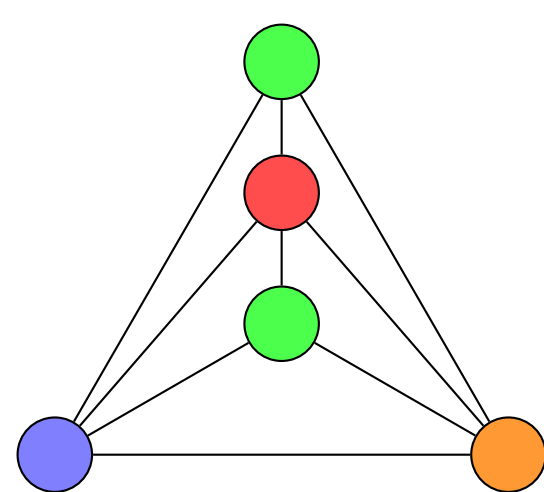


Figure 1: A graph colored with the proper coloring.

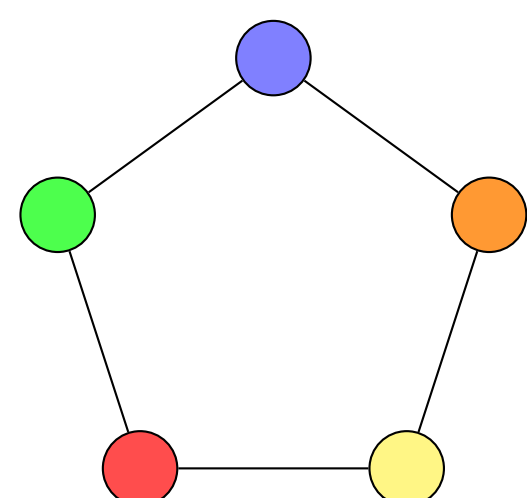


Figure 2: The graph C_5 colored with odd coloring.

- Odd coloring
- Proper coloring where every vertex has at least **one color** appearing an **odd** number of times in its neighborhood.

- Conflict-free (CF) coloring
- Every vertex has at least one **color** occurring **only once** in its neighborhood.
- Four versions:** proper/improper and open/closed neighborhoods

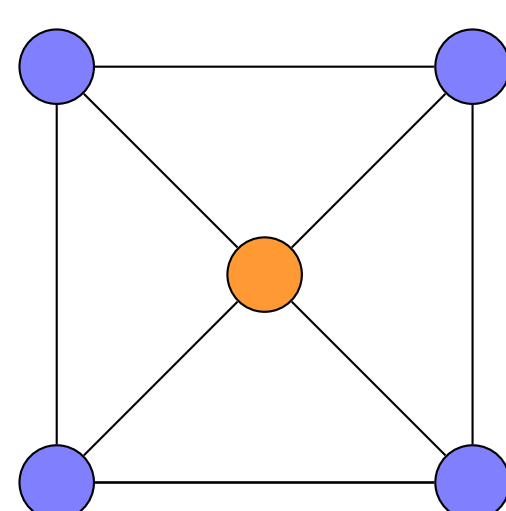


Figure 3: A graph colored with improper CF with respect to closed neighborhoods coloring.

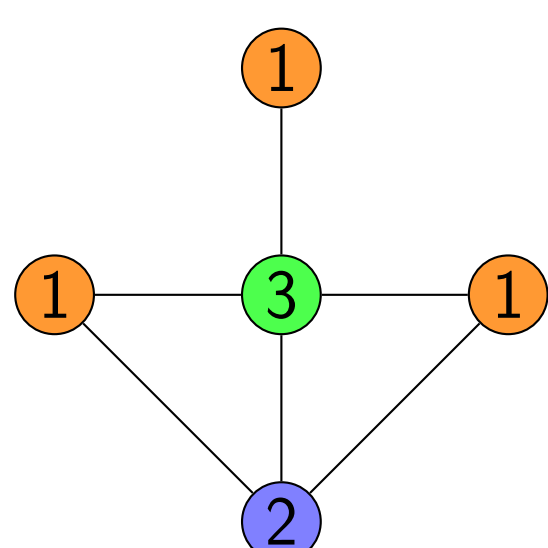


Figure 4: A colored graph for the proper UM with respect to open neighborhoods coloring.

- Unique-maximum (UM) coloring
- The **maximum color** in the neighborhood of a vertex occurs **exactly once**.
- Four versions:** proper/improper and open/closed neighborhoods

Chromatic numbers

Definition

Chromatic number $\chi_K(\mathcal{P})$ for coloring K , is the minimum needed number of colors to color all planar graphs $G \in \mathcal{P}$ according to K .

Chromatic numbers for these coloring variants are not always known. **Conjectures** on the chromatic numbers are tested by exhaustively coloring graphs.

Algorithm

Algorithm 1 describes a **high-level overview** of the designed algorithm. This is a **backtracking** algorithm using a multitude of optimizations (e.g., *second order neighborhood*, *parallelization*) to perform as efficiently as possible while preserving **expandability**.

Algorithm 1 search chromatic number

Parameters: The coloring K and graph G

Output: $\chi_K(G)$

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1: for max_c from 1 to MAX do
2:   while not all vertices are colored do
3:      $v \leftarrow \text{get\_optimal\_vertex}(G)$ 
4:      $\text{assign\_next\_possible\_color}(v, \text{max\_c})$ 
5:      $\text{update\_neighbors}(G, v)$ 
6:     if coloring of  $G$  is not possible then
7:        $\text{remove\_color}(v)$ 
8:     end if
9:   end while
10: end for
    
```

Results & Conclusion

Computational evidence for conjectures

Conjectures for planar triangulations with number of vertices $n \leq 20$ are correct.

- First **open-source** algorithm that colors graphs for these variants.
- Allows for searching of **gadgets**, such as the one shown in Figure 4, to construct graphs with high chromatic numbers.
- It only performs around 1.4 times worse than alternatives for the proper coloring. Tools for other colorings do not exist.